TECHNICAL WHITE PAPER · GOVERNMENT / DEFENCE BRIEFING

Federated Sovereign Starlink Model

A reference concept for partner-nation sovereignty over a Starlink-class LEO network: a five-level nationalization spectrum, a sovereign national ops center, and a federation layer

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CONCEPT, NOT AN OFFERING. This paper describes a hypothetical model. No public source shows SpaceX offering nationalized Starlink satellites, or transferring satellite command authority, to any foreign government. The closest public precedents are US-only: the government-owned MILNET / Space Data Network (SDN) Backbone and the NRO’s proliferated constellation. All partner nations, satellite counts, levels, sites, capacity shares, routing results and timelines in this paper are illustrative. No government, agency or company has endorsed or reviewed it.

Contents

1. Executive Summary....3
2. Background and Problem Statement: The Sovereignty Gap....3
3. Current Landscape....4
4. The Nationalization Spectrum (L0–L4)....5
5. Reference Architecture....7
6. The Federation Model and Illustrative Partners....11
7. Sovereignty Controls....12
8. Routing and Coverage Trade-offs....13
9. Risks, Feasibility and Policy Considerations....14
10. Recommendations and Phased Roadmap....15
Annex A. Assumptions and Illustrative Data....16
Annex B. Fact Status: Real versus Illustrative....17
References....18

Figures

Figure 1. L0–L4 control matrix....7
Figure 2. Five-layer reference architecture....8
Figure 3. SNOC block diagram....10
Figure 4. Illustrative dedicated satellites and reserved capacity share by partner (model figures, not real programmes)....12
Figure 5. Indicative phased roadmap....16

How to read this paper

Sourced facts carry bracketed reference numbers, e.g. [5], that resolve to the References list. Figures and statements marked **(illustrative)** are model assumptions and are not claims about any real programme, government or contract. Annex B lists every substantive claim with its status. Times and dates are as reported by the cited source.

1. Executive Summary

Governments increasingly depend on commercial low-Earth-orbit (LEO) broadband for defence, diplomatic and crisis communications, but they do not control the networks they depend on. Public episodes in Ukraine in 2022 and October 2026 [15][16] showed that the terms, coverage and performance of service delivered over Starlink can be decided outside the user government. This paper sets out a **Federated Sovereign Starlink Model**: a reference concept in which partner nations take progressively more control of a slice of a Starlink-class network, up to nation-owned satellites under national command, each operated from a **sovereign national ops center (SNOC)** that interfaces with SpaceX core operations and a multinational **federation layer**.

1.1 Key findings

- **Nationalization is a ladder, not a switch.** Five levels (L0–L4) move one control at a time from the operator to the nation: first where traffic lands, then who owns the satellites, finally who holds the command keys.
- **L0–L1 exist today; L2 is partly required by regulators; L3 has a US-only precedent; L4 has no public precedent.** India's 2025 licence conditions already mandate in-country gateways and local lawful intercept [18][19]; the US MILNET / SDN Backbone was reported as government-owned and contractor-operated with 480-plus Starshield satellites [4][26]; no public source shows SpaceX offering L4 to anyone.
- **A higher service tier does not buy sovereignty.** Musk has said limits on Starshield speed and usage "are decided by the US military and ultimately POTUS" [16]. Partners that need assured access need contractual no-cutoff terms, national command keys (L4), or both.
- **Small sovereign fleets cannot close routes alone.** In the model's illustrative routing, a 24-satellite national fleet often has no end-to-end laser path under sovereign-only routing; federated transit across partners closes many of these gaps. This is the central sovereignty-versus-coverage trade-off.
- **SpaceX retains launch, manufacturing, the software supply chain and global route computation at every level.** Dependence is reduced and made contractual, not removed. A non-SpaceX fallback bearer such as IRIS² (348 satellites, first launches planned for 2029 [21][22]) is part of any credible exit strategy.

1.2 Recommendations

1. **Start at L2.** Stand up national gateways, PoPs, lawful-intercept enclaves and national inner encryption now; this is achievable within existing commercial and regulatory practice.
2. **Negotiate a master agreement before buying satellites.** Assured access, no unilateral cutoff of national assets, escrowed ground software, step-in and buy-out rights should precede any L3 commitment.
3. **Form the federation early.** A small partner charter for reciprocal transit, capacity accounting, shared space situational awareness (SSA) and cyber indicators is what makes small national fleets useful.
4. **Treat L3 as the realistic ceiling for the next five years and L4 as an option.** L3 requires US government approval for a foreign customer and export-control clearance; L4 also needs SpaceX willingness, TT&C and crypto integration and an ITU filing strategy.
5. **Fund a second bearer.** Pair any Starlink-based sovereignty arrangement with IRIS², GEO milsatcom or another non-SpaceX path, so that exit is credible.

2. Background and Problem Statement: The Sovereignty Gap

LEO mega-constellations deliver global, low-latency broadband at a scale national programmes cannot quickly match. SpaceX's Starlink is the largest: third-party trackers count roughly 11,000 Starlink satellites

in orbit as of late October 2026 (an estimate, not an official figure), and on 9 January 2026 the FCC authorised a second 7,500-satellite tranche, bringing the authorised Gen2 constellation to 15,000 satellites [11][12]. The network's satellites are linked by optical inter-satellite lasers, three per satellite at up to 200 Gbps [8], and it lands traffic through a large gateway network, with more than 100 US gateway sites and over 1,500 antennas [10].

For a government user, the difficulty is not capability but control. In the commercial and government tiers, the operator owns and commands the satellites, computes routes, decides where terminals work and is subject to US law and policy. The **sovereignty gap** is the distance between the operational dependence a government accepts and the control it actually holds. It has four parts:

- **Access**: whether service can be restricted, geofenced or degraded by a third party, or for reasons the user cannot see.
- **Data**: where traffic lands, where it is decrypted, and whose law governs intercept, metadata and logs.
- **Command**: who can command the satellites carrying national traffic and who holds the keys.
- **Continuity**: whether the nation can keep operating, or exit, if the commercial relationship or US policy changes.

Public episodes illustrate each part. In September 2022, Starlink coverage near Crimea was not enabled for a Ukrainian operation; Musk said he refused a Kyiv request, and accounts differ on whether service was cut or never enabled [15]. In October 2026, Ukraine reported that military drones lost satellite connectivity above about 160 km/h (100 mph) for around ten days; the cause and responsibility are disputed, and the Pentagon Inspector General declined a bipartisan Senate request to investigate [16]. Separately, Italy's reported talks on a Starlink government contract stalled in 2025 amid dependence concerns [20][27].

2.1 Problem statement

How can a partner nation obtain assured access, data sovereignty, and (where needed) command authority over the part of a Starlink-class network that carries its national traffic, without building a national constellation from scratch, and while keeping the coverage benefits of a shared global mesh?

2.2 Scope and method

This paper is a design concept. It uses public facts about Starlink, Starshield and related programmes, all cited, and an illustrative eight-nation federation to test the design. It does not describe any SpaceX product, offer or negotiation, and it does not assess classified capabilities.

3. Current Landscape

The following programmes and rules are the factual basis for the model. Each is drawn from public reporting or official sources.

Table 1. Public landscape: programmes, contracts and rules relevant to sovereign LEO service

Item	What is publicly known	Relevance to the model	Ref.
Starshield	SpaceX's government line: Earth observation, communications and hosted payloads; adds high-assurance cryptographic capability for classified payloads; laser terminals can be integrated on partner satellites.	Basis for L1 (government tier) and the satellite bus for L3/L4.	[1]
NRO proliferated architecture	Launches since May 2024; NROL-105 on 16 Jan 2026 was the 12th; hundreds of satellites planned, launches through 2029. Reported as Starshield-based; not officially confirmed.	Precedent for a government-owned constellation built on a commercial-derived bus.	[2][3]

Item	What is publicly known	Relevance to the model	Ref.
MILNET / SDN Backbone	Reported in 2025 as a government-owned, contractor-operated (GOCO) constellation of 480-plus Starshield satellites with a Space Force (Delta 8) mission director. Renamed SDN Backbone; SpaceX won a $2.29B firm-fixed-price OTA in May 2026, with a fully operational prototype due by end-2027. Space Systems Command described it as government owned and operated.	Closest precedent for L3. US only.	[4][5][6] [26]
Starlink laser mesh	Three lasers per satellite at up to 200 Gbps; about 9,000 lasers carrying over 42 PB/day (early 2024).	Enables policy-tagged routing and federated transit.	[8][9]
Gateways	More than 100 US gateway sites and over 1,500 antennas.	National gateways and PoPs are the L2 landing point.	[10]
FCC Gen2 grant (DA 26-36)	9 Jan 2026: second 7,500-satellite tranche, 15,000 Gen2 satellites authorised; US is the notifying administration.	Satellites in this model fly in licensed shells under US filings.	[11][12]
Starlink V3 / Starship	28 Sep 2026: Starship Flight 14 reached orbit for the first time and deployed 26 Starlink V3 satellites.	Larger, higher-capacity satellites change the economics of dedicated fleets.	[13][14]
Ukraine 2022 / 2023 / 2026	2022 Crimea episode; June 2023 Pentagon contract for Starlink in Ukraine, reported to give DoD control of signal settings; Oct 2026 drone-speed restriction episode.	Motivates no-cutoff terms and national command.	[15][16] [17]
India licence (2025)	Indian gateways only; no copying or decryption of Indian data abroad; local lawful intercept.	Regulatory precedent for L2 (regulation, not ownership).	[18][19]
Italy	Reported ~€1.5B, five-year government Starlink deal stalled in March 2025 over political and dependence concerns.	Demand exists; dependence is the obstacle.	[20][27]
IRIS²	EU secure connectivity constellation: 348 satellites (330 LEO, 18 MEO) after the Aug 2026 implementation agreement; first launches 2029, services from 2030.	Non-SpaceX fallback bearer and exit path.	[21][22]
South Africa	Licensees must be 30% owned by historically disadvantaged groups; SpaceX called this a significant barrier. A Dec 2025 policy direction allows equity-equivalent programmes, but as of Oct 2026 Starlink still lacked market access.	Landing rights are national and can block entry entirely.	[23][24]
Export control	Military spacecraft and related data are generally ITAR (USML Cat. XV); most commercial comsats moved to EAR 9x515 in 2014.	Shapes what a foreign nation can receive at L3/L4.	[25]

Gap in the record. Every public example of a government owning or directing Starshield-class satellites is a US government programme. No public source shows a foreign government owning Starlink or Starshield satellites, or holding their command keys.

4. The Nationalization Spectrum (L0–L4)

The model defines five levels. Each moves one more control from SpaceX to the nation. The levels are cumulative: L3 includes the L2 sovereign partition, and L4 includes L3's dedicated satellites.

Table 2. The five levels of nationalization

Level	What it means	Owns sats	Commands	Traffic lands	Public status	Precedent
L0 Commercial service	Subscription on the shared Starlink network	SpaceX	SpaceX	SpaceX gateways and PoPs	Exists publicly	Starlink consumer, business and government plans
L1 Government service tier	Starshield terminals and contracted service terms	SpaceX	SpaceX, with contract terms	SpaceX network, extra encryption	Exists publicly	Starshield comms [1]; 2023 Pentagon deal for Ukraine reportedly gave DoD control of signal settings [17]
L2 Sovereign partition	National gateway, PoP, lawful intercept and data residency on shared satellites	SpaceX	SpaceX; nation sets slice policy	In-country gateway and PoP only	Partial precedent	India's 2025 licence conditions (regulation, not ownership) [18][19]
L3 Dedicated satellites, GOCO	Nation-funded Starshield-class satellites operated by SpaceX crews under a national mission director	Nation	SpaceX crews, tasked by nation	National gateway; mesh by policy	Partial precedent	US MILNET / SDN Backbone; NRO proliferated constellation [2][4][5]. No public foreign equivalent
L4 Full national ownership and command	Nation owns satellites, runs TT&C and holds command keys; SpaceX supplies bus, launch and mesh access	Nation	Nation	National gateway; sovereign mesh island possible	Proposed concept	No public source shows SpaceX offering this to any government

4.1 What changes at each step

- **L0 → L1:** contractual terms and higher-assurance terminals and crypto; the operator still decides coverage and policy.
- **L1 → L2:** data sovereignty. All national traffic lands, exits and is intercepted in-country; the nation sets slice policy, while the satellites remain shared.
- **L2 → L3:** asset ownership. The nation funds and owns dedicated satellites; SpaceX crews fly them at the tempo set by a national mission director, mirroring the GOCO pattern reported for MILNET [4].
- **L3 → L4:** command authority. The nation uplinks from its own TT&C station with its own keys and can isolate its satellites into a sovereign routing island.

4.2 Control matrix

Figure 1 consolidates who holds each function at every level. Rows for L2–L4 follow the source model's split-of-control design; L0–L1 cells are inferred from the level definitions and are marked with an asterisk where a contract or Starshield feature shares control.

	L0 Commercial	L1 Gov. tier	L2 Sovereign partition	L3 Dedicated GOCO	L4 National command
Owns satellites	SpaceX	SpaceX	SpaceX	Nation	Nation
Commands satellites	SpaceX	SpaceX*	SpaceX	SpaceX (nation-tasked)	Nation
Traffic lands	SpaceX GW/PoP	SpaceX network	National GW/PoP	National GW/PoP	National GW/PoP
Launch, build, replenish	SpaceX	SpaceX	SpaceX	SpaceX	SpaceX
Safety of flight & disposal	SpaceX	SpaceX	SpaceX	SpaceX	Shared
Bus software updates	SpaceX	SpaceX	SpaceX	Shared	Nation
Satellite command (TT&C)	SpaceX	SpaceX	SpaceX	SpaceX	Nation
Beam / coverage policy	SpaceX	Shared*	Shared	Nation	Nation
Laser-mesh routing	SpaceX	SpaceX	SpaceX	Shared	Shared
Terminal authorisation	SpaceX	Shared*	Shared	Nation	Nation
Inner encryption & keys	User	Shared*	Nation	Nation	Nation
Gateway, PoP, lawful intercept	SpaceX	SpaceX	Nation	Nation	Nation
ITU filing	US filing	US filing	US filing	US filing	US filing
Cross-partner routing	n/a	n/a	Federation	Federation	Federation

SpaceX · Nation · Shared · Federation · US filing · User / customer

Figure 1. L0–L4 control matrix. L2–L4 per the model's split of control (Section 5.3); L0–L1 inferred from level definitions. * = shared via contract terms or Starshield features. Design assumptions, not a description of any existing contract.

5. Reference Architecture

5.1 Five-layer architecture

The architecture has five layers, each with components that are nation-controlled, federated (shared) or SpaceX-retained. The colour coding in Figure 2 is the core of the design: sovereignty is achieved by moving specific components, not by moving the whole system.

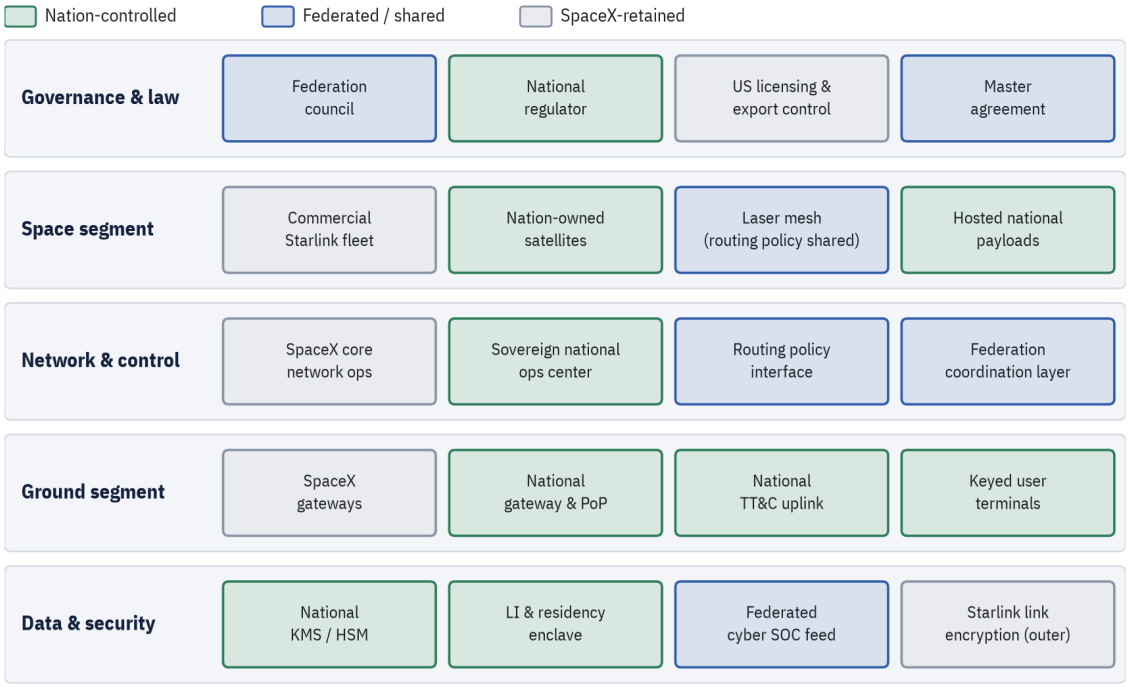


Figure 2. Five-layer reference architecture. Green: nation-controlled. Blue: federated / shared. Grey: SpaceX-retained.

Table 3. Architecture components by layer and control

Layer	Component	Control	Role
Governance & law	Federation council	Federated	Partners set cross-routing, capacity-sharing and crisis rules; each nation keeps a veto over its own assets
	National regulator	Nation	Landing rights, gateway and terminal licences, lawful-intercept and localisation conditions
	US licensing & export control	SpaceX / US	FCC licence and ITU filings under the US administration; ITAR/EAR govern transfers
	Master agreement	Federated	Assured access, no unilateral cutoff of national assets, escrow and step-in rights
Space segment	Commercial Starlink fleet	SpaceX	Shared satellites in licensed shells; carry partitioned national traffic as ciphertext
	Nation-owned satellites	Nation	Starshield-class buses funded and owned by a nation (L3/L4), flown in licensed shells
	Laser mesh	Federated	Optical inter-satellite links; routing policy is shared
	Hosted national payloads	Nation	Crypto, sensing or relay payloads under exclusive national control
Network & control	SpaceX core network ops	SpaceX	Fleet health, safety of flight, global route computation, software supply chain
	Sovereign national ops center	Nation	National mission direction or TT&C, key custody, slice policy, cyber SOC
	Routing policy interface	Federated	Nations publish route constraints that SpaceX's router must honour
	Federation coordination layer	Federated	Brokers cross-partner capacity and routing; shares SSA and cyber indicators

Layer	Component	Control	Role
Ground segment	SpaceX gateways	SpaceX	Global gateway network for non-sovereign traffic
	National gateway & PoP	Nation	In-country landing point and internet exit for all national traffic
	National TT&C uplink	Nation	Command uplink (L4) or mission-director link (L3)
	Keyed user terminals	Nation	Starshield-class terminals with national crypto and authorisation lists
Data & security	National KMS / HSM	Nation	Nation-held root keys for inner encryption and command authentication
	LI & residency enclave	Nation	Lawful intercept, metadata and logs stay in-country under national law
	Federated cyber SOC feed	Federated	Shared indicators and incident coordination
	Starlink link encryption	SpaceX	SpaceX's own user-data encryption, the outer layer

5.2 The sovereign national ops center (SNOC)

Each partner runs a SNOC in-country. It is the single point where national authority meets the shared network. Its six components are:

Table 4. SNOC components

Component	Function
Command authority (TT&C / mission direction)	L3: a national mission director tasks SpaceX crews, as reported for MILNET's Delta 8 director [4]. L4: national uplink and command keys for owned satellites.
Key custody	National HSMs hold root keys for inner traffic encryption, terminal keying and satellite command authentication; SpaceX never holds them.
National gateway and PoP	All national traffic lands and exits in-country; DNS and internet exit are local.
Lawful-intercept and data-residency enclave	Intercept, metadata, terminal location and logs stay in-country under national law.
Cyber SOC	Monitors terminals, gateway and control interfaces; exchanges indicators with SpaceX and the federation SOC.
Policy desk	Publishes routing constraints, coverage rules, terminal allow-lists and crisis priorities to the interfaces.

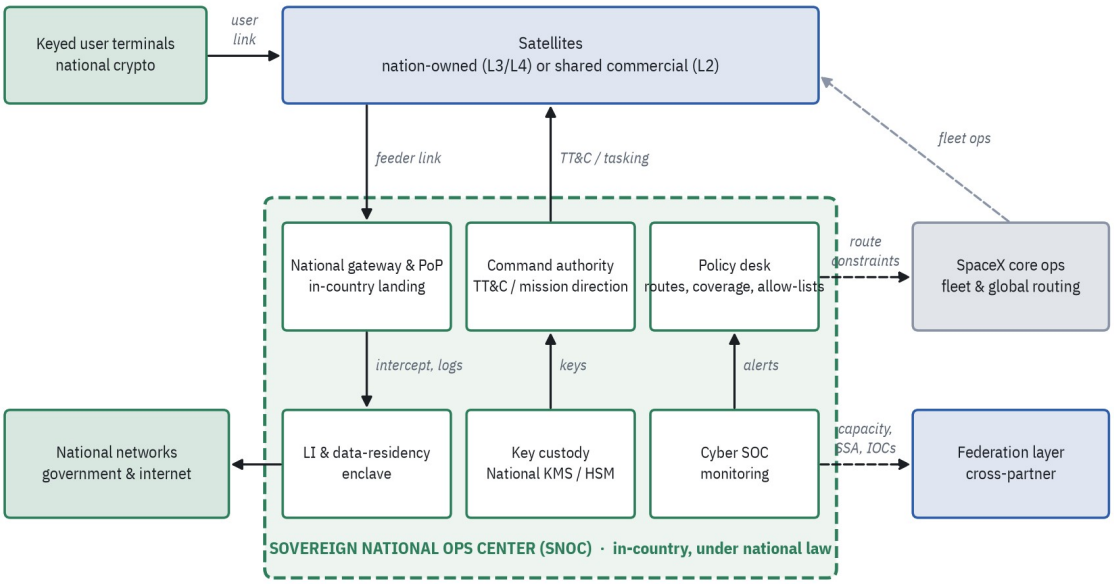


Figure 3. SNOC block diagram. Solid arrows: national data and command paths. Dashed: interfaces to SpaceX core ops and the federation layer.

5.3 Split of control with SpaceX core ops and the federation layer

Table 5. Split of control by level (design of the model)

Function	L2	L3	L4	How it works
Launch, manufacturing, replenishment	SpaceX	SpaceX	SpaceX	SpaceX builds and launches at every level; the nation's leverage is contractual
Safety of flight and disposal	SpaceX	SpaceX	Shared	Starlink satellites manoeuvre autonomously to avoid collisions; at L4 the nation commands, with SpaceX screening kept on
Bus software updates	SpaceX	Shared	Nation	SpaceX signs builds; at L3 the nation approves updates to its satellites, at L4 it decides when to load them
Satellite command (TT&C)	SpaceX	SpaceX	Nation	L3 mirrors MILNET: contractor crews execute at a tempo set by a national mission director. L4 uplinks from a national station with national keys
Beam and coverage policy for national users	Shared	Nation	Nation	The nation sets where its terminals work; no third party can geofence national terminals on national satellites
Laser-mesh routing	SpaceX	Shared	Shared	SpaceX's router computes paths; nations supply binding constraints. At L4 a nation can switch its satellites to a sovereign-only island
Terminal authorisation	Shared	Nation	Nation	National allow-lists and terminal keys are loaded and revoked by the SNOC
Inner encryption and keys	Nation	Nation	Nation	National crypto rides inside Starlink's own encryption at every level
Gateway, PoP and lawful intercept	Nation	Nation	Nation	In-country landing, internet exit, intercept and logs, as India's licence conditions already require
ITU filing for the satellites	US filing	US filing	US filing	Starlink is filed by the US administration; a national filing for L4 is possible in principle but needs new coordination and milestones
Cross-partner routing and capacity	Federation	Federation	Federation	The federation layer brokers use of other partners' satellites under reciprocal agreements

What SpaceX retains at every level

- **Launch, manufacturing and the software supply chain:** the nation depends on SpaceX for replenishment and signed builds.
- **Global route computation:** the laser mesh is one network; nations constrain routes but do not run the global router.
- **Commercial fleet and global gateways:** shared satellites stay SpaceX-owned and SpaceX-commanded.
- **US regulatory exposure:** FCC licence, US ITU filings, and US export and sanctions law apply to SpaceX's actions.

6. The Federation Model and Illustrative Partners

The federation is a group of partner nations that agree to carry each other's traffic across their dedicated satellites, broker shared capacity, and exchange SSA and cyber indicators. It adds a layer between individual SNOCs and SpaceX core operations, so that small national fleets can borrow coverage without giving up control of their own assets.

> **Illustrative data.** The eight partners below are hypothetical. Their levels, satellite counts, sites and capacity reservations are example figures chosen to exercise the design. Only the orbital shells are real FCC-licensed parameters [11]. Nothing here implies any of these governments has sought, discussed or been offered such an arrangement.

Table 6. Illustrative federation partners (all figures illustrative)

Nation	Level	Sats	Shells (illustrative allocation)	Ops center	Gateway & PoP	Reserved
United States	L3 Dedicated GOCO	120	Gen2 480 km / 53° ×72; Gen2 360 km / 96.9° ×48	Colorado region	Arizona region	30%
United Kingdom	L4 National command	48	Gen2 480 km / 53° ×48	Southern England	Scotland	12%
Japan	L3 Dedicated GOCO	48	Gen2 485 km / 43° ×32; Gen2 475 km / 32° ×16	Tokyo region	Hokkaido	10%
Australia	L3 Dedicated GOCO	40	Gen2 485 km / 43° ×24; Gen2 360 km / 96.9° ×16	Canberra region	Western Australia	9%
Canada	L3 Dedicated GOCO	36	Gen2 360 km / 96.9° ×36	Ottawa region	Yellowknife region	8%
Italy	L4 National command	36	Gen2 485 km / 43° ×36	Rome region	Southern Italy	8%
Poland	L3 Dedicated GOCO	24	Gen2 480 km / 53° ×24	Warsaw region	Kraków region	6%
India	L2 Sovereign partition	0	Shared satellites only	Delhi region	Mumbai region	17%
Total		352				100%

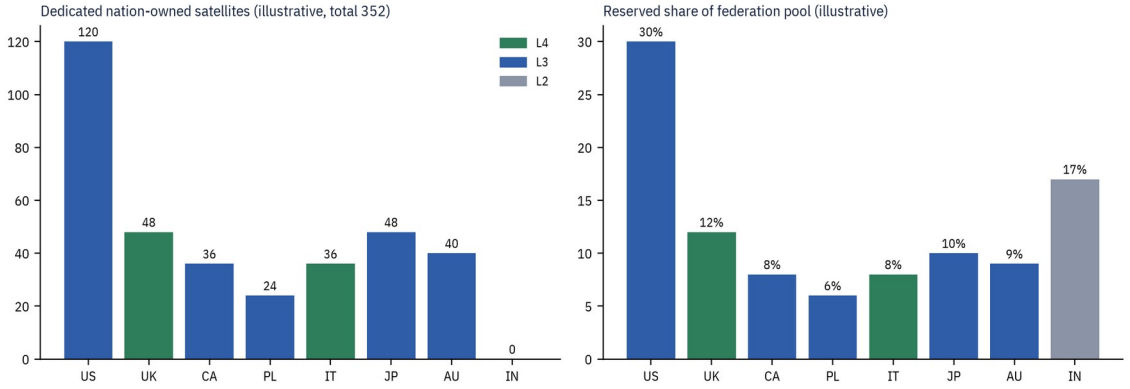


Figure 4. Illustrative dedicated satellites and reserved capacity share by partner (model figures, not real programmes).

6.1 Roles of the illustrative partners

- **United States (L3, anchor):** mirrors the public GOCO pattern; largest fleet and capacity reservation.
- **United Kingdom and Italy (L4):** full national ownership and command of dedicated satellites, in the 53° and 43° shells respectively.
- **Canada (L3):** polar shell for Arctic coverage. **Australia (L3):** southern-hemisphere and polar access. **Japan (L3):** mid-latitude and low-inclination shells for the western Pacific.
- **Poland (L3):** smallest dedicated fleet; leans on federated routing.
- **India (L2):** sovereign partition on shared satellites, with no dedicated satellites; consistent with the type of regulatory conditions India already applies [18][19].

6.2 Federation functions

Table 7. Federation layer functions

Function	Description
Reciprocal transit	Partner-owned satellites carry each other's traffic as ciphertext under transit agreements
Capacity brokering	Federation pool shares reserved per partner (illustrative shares above) and accounted per use
Crisis rules	Pre-agreed priorities and a veto for each nation over its own assets
Shared SSA and cyber indicators	Federated SOC feed and space situational awareness exchange with SpaceX
Landing-rights annexes	Reciprocal annexes so federated transit can land at partner gateways where agreed

7. Sovereignty Controls

Each control pairs a proposed mechanism with the public reality or precedent that motivates it.

Table 8. Sovereignty controls paired with real precedents

Control	Proposed mechanism	Public reality and precedent
Assured access, no remote cutoff	Contract bars unilateral geofencing, throttling or cutoff of national terminals on nation-owned satellites; L4 command keys make this technically enforceable	Sept 2022: Starlink coverage near Crimea was not enabled for a Ukrainian operation; accounts differ on whether it was cut [15]. Oct 2026: Ukraine reported drones losing connectivity above about 160 km/h for about 10 days; cause disputed [16]

Control	Proposed mechanism	Public reality and precedent
Crypto	National inner encryption and terminal keying from national HSMs, over Starlink's own encryption	SpaceX says Starlink encrypts user data; Starshield adds high-assurance crypto for classified payloads [1]
Data localisation and lawful intercept	In-country gateway, PoP, DNS, intercept and logs; no mirroring abroad	India's 2025 licence: Indian gateways only, no copying or decryption of Indian data abroad, local lawful intercept [18][19]
Export control / ITAR	Black-box integration: nation receives capability, not design data; government-to-government channel where required	Military spacecraft and related data are generally ITAR (USML Cat. XV); most commercial comsats moved to EAR 9x515 in 2014 [25]
ITU filings	L2–L3 stay under US filings; L4 may stay under them by agreement, or move to a national filing with new coordination	FCC Gen2 grant lists the US as notifying administration and ties deployment to ITU milestones [11]
Landing rights	Each nation licenses its own gateway and terminals; federation transit needs reciprocal landing-rights annexes	Granted country by country; e.g. South Africa's 30% ownership rule has kept Starlink out; reform still pending in Oct 2026 [23][24]
Exit and continuity	Escrowed ground software, step-in rights, buy-out terms, and a non-SpaceX bearer such as IRIS² or GEO	Italy's reported Starlink talks stalled in 2025 over dependence concerns [20][27]; IRIS² first launches planned for 2029 [21][22]

"All limits on Starshield speed and usage are decided by the US military and ultimately POTUS." — Elon Musk, as reported by Ars Technica, 29 October 2026 [16]

Even the government-grade line answers to US authorities. That is why foreign partners would want command keys (L4) or a contractual bar on cutoffs rather than simply a higher service tier. Note that Ars also reports that President Trump denied ordering the October 2026 restrictions and that the administration said it had no foreknowledge of them [16]; responsibility remains disputed.

8. Routing and Coverage Trade-offs

8.1 Laser-mesh routing handoff

National traffic crosses the shared mesh under policy tags. The handoff sequence is:

6. **Ingress.** A national terminal connects to the best satellite its policy allows; traffic is already under national inner encryption.
7. **Policy-tagged routing.** SpaceX's router computes paths that honour the nation's tags: allowed satellites, required landing gateway, regions to avoid.
8. **Boundary crossing.** When a path leaves nation-owned satellites for partner or commercial ones, the handoff is logged to the SNOC and carriers see only ciphertext.
9. **Federated transit.** Partner-owned satellites carry each other's traffic under reciprocal federation agreements, with capacity accounted per partner.
10. **Sovereign island.** In a crisis an L4 nation can restrict its satellites to route only among themselves and land only at its gateway, accepting coverage gaps.

8.2 Three routing policies

Table 9. Routing policies and the sovereignty-versus-coverage trade-off

Policy	Eligible satellites	Sovereignty	Coverage and resilience
Sovereign only	The nation's own dedicated satellites	Highest: no third-party carriage	Lowest: small fleets often lack an end-to-end path

Policy	Eligible satellites	Sovereignty	Coverage and resilience
Federated partners	All partners' dedicated satellites	High: carriage only by treaty partners, as ciphertext	Medium: many gaps closed by partner fleets
Full Starlink mesh	Dedicated plus commercial satellites	Lower: commercial carriage as ciphertext	Highest: global mesh

8.3 Model result (illustrative)

The source model simulates the illustrative fleets in real licensed shells and traces a route from a deployed terminal (e.g. Poland's in the eastern Mediterranean) to the national gateway using a simple greedy laser-hop router (assumed 5,000 km laser range and 1,300 km access range). Under **sovereign-only** routing, Poland's 24-satellite fleet frequently has no end-to-end path; switching to **federated partners** usually closes it. When no path exists, traffic must queue, fall back to a wider policy, or use another bearer.

> **Caveat.** The router is a simple greedy model built for illustration, not Starlink's real routing algorithm. Its outputs show the direction of the trade-off, not real performance, latency or availability.

8.4 Design implications

- Fleet size and shell diversity matter more than raw capacity for sovereign-only routing.
- More national gateways shorten the path to landing and reduce dependence on transit.
- Federation transit agreements are an engineering requirement for small fleets, not just a diplomatic nicety.
- A sovereign island should be a declared crisis mode with a published coverage cost, not the default.

9. Risks, Feasibility and Policy Considerations

9.1 Gains and costs

Table 10. Trade-offs

Gains	Costs
Sovereign control at a fraction of a national constellation's cost and time	Lasting dependence on one US company for launch, software and routing
Global coverage and laser backhaul from the shared mesh	Sovereign-only routing leaves large coverage gaps
Data stays in-country under national law	US law and policy still reach SpaceX's decisions
Federated transit covers small fleets' gaps	Hard integration of national crypto and TT&C into a proprietary system

9.2 Feasibility by level

Table 11. Feasibility and indicative timelines

Level	Feasibility	Main obstacle	Indicative timeline (illustrative)
L1 Government tier	High: in use	Contract terms on coverage and cutoff	Available now

Level	Feasibility	Main obstacle	Indicative timeline (illustrative)
L2 Sovereign partition	High: regulators already require parts of it	Gateway build-out and LI integration	1–2 years
L3 Dedicated GOCO	Medium: US precedent only	US approval for a foreign customer; export control	3–5 years
L4 National command	Low: no public offer	SpaceX willingness, ITAR, TT&C and crypto integration, ITU filing	5+ years, if ever

9.3 Risk register

Table 12. Risk register

Risk	Why it matters	Mitigation
Unilateral restriction	Public episodes in 2022 and 2026 [15][16]	L4 keys, no-cutoff clauses, a second bearer
Vendor lock-in	No second source for buses, launch or mesh access	Escrow, step-in rights, IRIS² or GEO fallback
US policy shifts	Export licences and contracts can change	Government-to-government agreements, long licence terms
Mesh gaps under sovereign routing	Small fleets cannot close end-to-end paths	Federated transit, larger fleets, more gateways
Cyber exposure at interfaces	Policy and TT&C interfaces are high-value targets	Zero-trust interfaces, national SOC, federated indicators
Spectrum and filing conflicts	National L4 filings need fresh coordination	Keep L4 under US filings by agreement

9.4 Policy considerations

- **US government consent.** L3 and L4 for foreign customers would likely need US government approval and export licensing; the only public GOCO precedent is a US programme [4][5].
- **Export-control posture.** The 2014 reform moved most commercial comsats to the EAR, but military spacecraft and related data generally remain ITAR-controlled [25]. Black-box integration limits what design data a nation receives.
- **Spectrum and ITU.** Satellites flown in Starlink shells operate under US filings and FCC milestones [11]; a national filing would require new coordination.
- **National law.** Landing rights, ownership rules and lawful-intercept regimes are set country by country, as India and South Africa show [18][19][23][24].
- **Alliance politics.** Reliance on a single US firm is itself a political variable, as the Italian case illustrates [20][27]; a European or allied second bearer reduces that exposure [21][22].

10. Recommendations and Phased Roadmap

10.1 Recommendations

11. **Build the sovereign partition first (L2).** National gateway and PoP, LI and residency enclave, national KMS/HSM and keyed terminals deliver most of the data-sovereignty benefit without new satellites.

12. **Make assured access contractual.** Negotiate a master agreement with no unilateral cutoff of national terminals on national assets, transparency on any restriction, escrowed ground software, step-in and buy-out rights.
13. **Charter the federation.** Agree reciprocal transit, capacity accounting, crisis priorities, SSA and cyber-indicator sharing among a small set of partners before fleets are bought.
14. **Pursue L3 through government-to-government channels.** Use the GOCO pattern with a national mission director; plan for US approval and export licensing as critical-path items.
15. **Hold L4 as an option, not a plan.** Define the technical prerequisites (national TT&C, command authentication, sovereign-island mode) and the ITU approach so the option can be exercised if SpaceX and US authorities agree.
16. **Insure the exit.** Fund a non-SpaceX bearer (IRIS², GEO milsatcom or others) and test failover regularly.

10.2 Phased roadmap (illustrative)

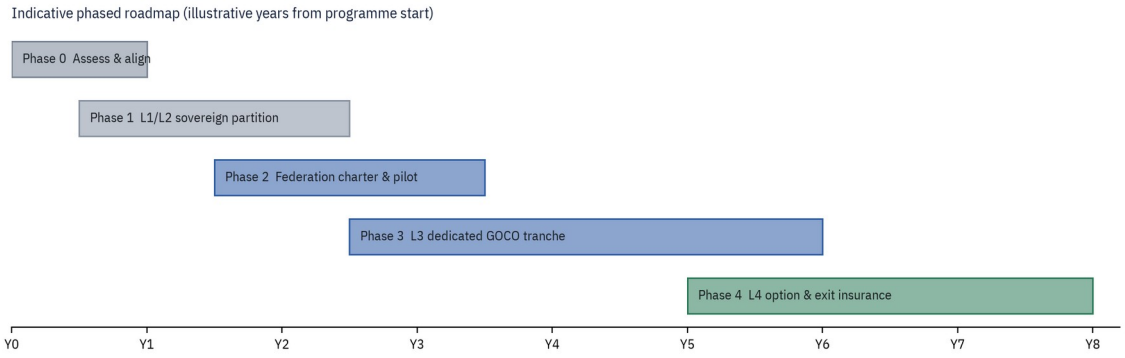


Figure 5. Indicative phased roadmap. Durations are illustrative and align with the feasibility timelines in Table 11.

Table 13. Roadmap phases (illustrative)

Phase	Objective	Key deliverables	Exit criteria
0 Assess & align	Define requirements and legal basis	Requirements, legal review, national policy decision	Mandate and funding
1 L1/L2 sovereign partition	Data sovereignty on shared satellites	National gateway & PoP, LI enclave, KMS/HSM, keyed terminals, SNOC core	Traffic lands only in-country; national keys in use
2 Federation charter & pilot	Reciprocal transit among partners	Charter, capacity accounting, SSA/SOC exchange, landing annexes	Pilot transit across at least two partners
3 L3 dedicated GOCO tranche	Nation-owned satellites under national mission direction	Master agreement, US approvals, first dedicated tranche, mission-director cell	National tasking of own satellites
4 L4 option & exit insurance	National command option; credible exit	TT&C station, command authentication, sovereign-island mode, second-bearer failover	Tested failover; L4 decision point

Annex A. Assumptions and Illustrative Data

The following are assumptions of the model. They are not claims about any real programme.

Table 14. Model assumptions

Item	Value used	Basis
Partner nations	US, UK, Canada, Poland, Italy, Japan, Australia, India	Hypothetical

Item	Value used	Basis
Levels per partner	L4: UK, Italy. L3: US, Canada, Poland, Japan, Australia. L2: India	Hypothetical
Dedicated satellites	352 total (US 120, UK 48, JP 48, AU 40, CA 36, IT 36, PL 24, IN 0)	Hypothetical
Capacity reservations	US 30%, IN 17%, UK 12%, JP 10%, AU 9%, CA 8%, IT 8%, PL 6% (sum 100%)	Hypothetical
Ops-center, gateway and terminal sites	Regional placeholders (e.g. Warsaw region, Kraków region, eastern Mediterranean)	Hypothetical
Federation layer host	Placeholder in Brussels area; could be distributed	Hypothetical
Laser ISL range (router)	5,000 km	Model assumption
Terminal/gateway access range (router)	1,300 km	Model assumption
Routing algorithm	Greedy nearest-to-gateway hop, up to 40 hops, Earth-occlusion check	Model assumption; not Starlink's router
Commercial fleet in simulation	Sampled at about 1 in 15, illustrative positions	Model assumption
Timelines	1–2 years (L2), 3–5 years (L3), 5+ years (L4)	Illustrative

Table 15**.** Orbital shells used in the model (shell parameters as cited in the source Page; allocation of national satellites to shells is illustrative)

Shell	Altitude	Inclination	Licence source
Gen1	550 km	53°	FCC 21-48
Gen1	570 km	70°	FCC 21-48
Gen1	560 km	97.6°	FCC 21-48
Gen2	480 km	53°	FCC DA 26-36
Gen2	485 km	43°	FCC DA 26-36
Gen2	475 km	32°	FCC DA 26-36
Gen2	360 km	96.9°	FCC DA 26-36
Gen2	340 km	53°	FCC DA 26-36

Annex B. Fact Status: Real versus Illustrative

Table 16**.** Fact status of claims in this paper

Claim	Status	Source
Starshield offers Earth observation, communications and hosted payloads; adds high-assurance crypto; laser terminals can go on partner satellites	Public	[1]
NRO proliferated constellation: launches since May 2024; NROL-105 (Jan 2026) the 12th; more through 2029; reported Starshield-based, not officially confirmed	Public	[2][3]
MILNET: GOCO, 480-plus Starshield satellites, Delta 8 mission director	Public (reported)	[4][26]
SDN Backbone (ex-MILNET): $2.29B award, prototype by end-2027	Public	[5][6][7]
Three lasers per satellite up to 200 Gbps; about 9,000 lasers and 42 PB/day (early 2024)	Public	[8][9]

Claim	Status	Source
More than 100 US gateway sites and 1,500 antennas	Public (from source Page; not re-checked)	[10]
FCC Gen2 grant: 15,000 Gen2 satellites authorised, 9 Jan 2026	Public	[11][12]
Gen2 shell parameters 340–485 km	Public (from source Page; not re-checked line by line)	[11]
Starship Flight 14 reached orbit and deployed 26 Starlink V3 satellites, 28 Sep 2026	Public	[13][14]
About 11,000 Starlink satellites in orbit, late Oct 2026	Estimate	Third-party trackers (per source Page)
Crimea 2022 and Ukraine Oct 2026 service episodes	Public; responsibility disputed	[15][16]
2023 Pentagon Starlink contract for Ukraine; DoD control of signal settings	Public (reported)	[17]
India's satcom security conditions (2025)	Public	[18][19]
Italy's Starlink talks stalled (2025)	Public	[20][27]
IRIS²: 348 satellites (330 LEO, 18 MEO), first launches 2029	Public	[21][22]
South Africa 30% ownership rule; Starlink not licensed as of Oct 2026	Public	[23][24]
ITAR USML Cat. XV / EAR 9x515 (2014 reform)	Public	[25]
SpaceX offering nationalized satellites or command to foreign governments	Not found	No public source
Partner nations, satellite counts, levels, sites, reservations, routing results, timelines	Illustrative	This model

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